Type Theory & Categorical Models: A Unified Approach

From Typelis to Logic to Closed Cartesian Categories

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Assumptions

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Syntax of -terms

- Variables, abstraction $\lambda x.\, M$, application $M\, N$

Subterms & Free variables

- Subterm relation: $N \preceq M$ if N occurs in M
- Free variables $\mathrm{FV}(M)$
- Combinators, when $\mathrm{FV}(M) = \varnothing$

Substitution

- Capture-avoiding substitution $M[x := N]$

α-equivalence

- Renaming of bound variables: $\lambda x.M \equiv_\alpha \lambda y.M[x := y]$ if $y \notin \mathrm{FV}(M)$
- Formal inference rules: reflexivity, congruence, renaming under binder

β-reduction

- Single-step: $(\lambda x.M)N \rightsquigarrow_\beta M[x := N]$
- Multi-step: $\rightsquigarrow_\beta^*$ (reflexive, transitive closure)
- Redex = reducible expression; normal form = no redex

η-equivalence

- Extensionality: $\lambda x.\, M\, x =_\eta M$ if $x \notin \mathrm{FV}(M)$

Church–Rosser

- Confluence of β-reduction:

$$M \rightsquigarrow_\beta^* N_1 \text{ and } M \rightsquigarrow_\beta^* N_2 \text{ then } \exists P.\ N_1 \rightsquigarrow_\beta^* P \wedge N_2 \rightsquigarrow_\beta^* P$$

- Proof idea: parallel reduction / diamond property

Normalization & Examples

- Strongly normalizing, weakly normalizing, non-normalizing
- $\lambda x.x$ (normal), Ω (non-normalizing), $(\lambda x.y)\Omega$ (reduces to y)

Encodings & Fixed-points

- Church numerals: $\mathrm{ZERO}, \mathrm{SUCC}, \mathrm{ADD}, \mathrm{MUL}, \mathrm{PRED}$
- Booleans: TRUE, FALSE, IF
- Pairs: PAIR, FST, SND
- Lists: NIL, CONS, HEAD, TAIL
- Fixed-point combinators: Y, Z, Θ

Typing

Minimal Typing

Types

$$\mathscr{T} ::= \mathscr{B} \mid \mathscr{T} \to \mathscr{T}$$

$$\lambda x : \sigma . x \vdash \lambda x : \sigma . x : \sigma \to \sigma$$

$$\lambda x . x \vdash \lambda x . x : \sigma \to \sigma$$

Church–Rosser ✓ Strongly Normalizing ✓ Weakly Normalizing ✓

Type Extensions

Product Types:	$A \times B \to A$	$A \times B \to B$
Sum Types:	$A \to A + B$	$B \to A + B$

In terms of β-reduction they are all confluent.

```
-- Product Type: A x B
data Pair a b = Pair a b
-- Projections
pi1 :: Pair a b -> a
pi1 (Pair x _) = x
pi2 :: Pair a b -> b
pi2 (Pair _ y) = y
```

```
-- Sum type: A + B
data Sum a b = InL a | InR b
-- Injections
inl :: a -> Sum a b
inl = InL
inr :: b -> Sum a b
inr = InR
```

Extended Typing – Products

Product Types

$$\mathscr{T}^{\times} ::= \mathscr{B} \mid \mathscr{T}^{\times} \to \mathscr{T}^{\times} \mid \mathscr{T}^{\times} \times \mathscr{T}^{\times} \mid 1$$

Product Terms

$$\Lambda^{\times} ::= x \mid \lambda x : \mathscr{T}.\ \Lambda^{\times} \mid \Lambda^{\times}\ \Lambda^{\times} \mid \langle \Lambda^{\times}, \Lambda^{\times} \rangle \mid \pi_1 \Lambda^{\times} \mid \pi_2 \Lambda^{\times} \mid *$$

Church–Rosser $\beta\eta$ ✗ Strongly Normalizing ✓ Weakly Normalizing ✓

Extended Typing – Sums

Sum Types

$$\mathscr{T}^+ ::= \mathscr{B} \mid \mathscr{T}^+ \to \mathscr{T}^+ \mid \mathscr{T}^+ + \mathscr{T}^+ \mid 0$$

Sum Terms

$$\Lambda^+ ::= x \mid \lambda x : \mathscr{T}.\ \Lambda^+ \mid \Lambda^+\,\Lambda^+ \mid \iota_1\Lambda^+ \mid \iota_2\Lambda^+ \mid \mathsf{case}\ \Lambda^+\ \mathsf{of}\ \iota_1 x \Rightarrow \Lambda^+ \mid \iota_2 y \Rightarrow \Lambda^+ \mid \Box_A\,\Lambda^+$$

Church–Rosser $\beta\eta$ ✗ Strongly Normalizing ✓ Weakly Normalizing ✓

Extended Typing – Products and Sums

Product Sum Types

$$\mathscr{T}^{\times+} ::= \mathscr{B} \mid \mathscr{T}^{\times+} \to \mathscr{T}^{\times+} \mid \mathscr{T}^{\times+} \times \mathscr{T}^{\times+} \mid \mathscr{T}^{\times+} + \mathscr{T}^{\times+} \mid 1 \mid 0$$

Product Sum Terms

$$\Lambda^{\times+} ::= \Lambda^{\times} \cup \Lambda^{+}$$

Supplementary Material

- **Church-Rosser Property (Untyped λ-Calculus)**:
  - Confluence of reduction: if $M \rightsquigarrow_\beta^* N_1$ and $M \rightsquigarrow_\beta^* N_2$, there exists P with $N_1 \rightsquigarrow_\beta^* P$ and $N_2 \rightsquigarrow_\beta^* P$
  - Proven using **parallel $\beta\eta$-reduction**
- **Weak Normalization (Typed λ-Calculus with Products)**:
  - Every typable term eventually reduces to a normal form
  - **Degree** is used as a measure of term complexity to structure the induction
  - Extends standard normalization proofs to include **product types**

Curry Howard

Intuitionistic Logic

- Proofs must be **constructive**:
  - To prove $A \to B$, one must give a method that transforms any proof of A into a proof of B.
  - To prove $\exists x. P(x)$, one must provide a concrete witness a together with a proof of $P(a)$.
- **Excluded Middle** does not hold in general: $\nvdash A \vee \neg A$ unless one can actually prove one of the disjuncts.
- Proofs by contradiction are not allowed.

The Isomorphism

Intuitionistic Logic	**Typed λ-Calculus**
Proposition A	Type A
Proof of A	Term of type A
Implication $A \to B$	Function type $A \to B$
Conjunction $A \wedge B$	Product type $A \times B$
Disjunction $A \vee B$	Sum type $A + B$
$\top$	1
$\bot$	0
Proof normalization	β-reduction

Example: Modus Ponens

Modus Ponens

$$\frac{P \to Q \quad P}{Q}\,\mathrm{MP}$$

$$\begin{aligned}
&(\lambda f : P \to Q.\lambda x : P.f\,x)\ f : P \to Q\ x : P && \vdash (P \to Q) \to P \to Q \\
&\rightsquigarrow_\beta\ (\lambda x : P.f\,x)\ x : P && \vdash P \to Q \\
&\rightsquigarrow_\beta\ f\,x && \vdash Q
\end{aligned}$$

Closed Cartesian Categories

What is Category Theory

- Category theory is a general theory of mathematical structures and their relations
- The focus of study are relations among objects, and not the objects themselves

Definition of Category

Category

A **category** $\mathscr{C}$ consists of:

- A collection of **objects**: $\mathrm{Ob}(\mathscr{C})$
- For each pair of objects $A, B \in \mathrm{Ob}(\mathscr{C})$, a set of arrows $\mathrm{Hom}_{\mathscr{C}}(A, B)$
- For each $A \in \mathrm{Ob}(\mathscr{C})$, an **identity morphism** $\mathrm{id}_A \in \mathrm{Hom}_{\mathscr{C}}(A, A)$
- For each $f \in \mathrm{Hom}_{\mathscr{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathscr{C}}(B, C)$, a **composition** $g \circ f \in \mathrm{Hom}_{\mathscr{C}}(A, C)$

Axioms

These must satisfy:

- **Associativity:** $(h \circ g) \circ f = h \circ (g \circ f)$.
- **Identity:** $\mathrm{id}_B \circ f = f = f \circ \mathrm{id}_A$.

Example: Preorder as a Category

$$
\begin{tikzcd}
a \arrow[r, "f"] \arrow[rr, bend left=50, "g \circ f"] \arrow[rrr, bend left=70, "(h \circ g) \circ f = h \circ (g \circ f) = h \circ g \circ f"] & b \arrow[r, "g"] \arrow[rr, bend right=50, "h \circ g"'] & c \arrow[r, "h"] & d
\end{tikzcd}
$$

Example: Preorder as a Category

$$
\begin{tikzcd}
a \arrow[r, "\leq"] \arrow[rr, bend left, "\leq"] \arrow[rrr, bend left=50, "\leq"] & b \arrow[r, "\leq"] \arrow[rr, bend right, "\leq"'] & c \arrow[r, "\leq"] & d
\end{tikzcd}
$$

Commutative Diagrams

Definition

A diagram in a category $\mathscr{C}$ is said to **commute** if, for any two objects connected by more than one path of morphisms, all paths compose to the same morphism.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "h"'] & B \arrow[d, "g"] \\ & C \end{tikzcd}$$

This triangle commutes if:

$$g \circ f = h$$

meaning the two ways of going from A to C are equal. Commutativity ensures consistency of compositions.

Injectivity via Arrows

Injective Function

A function $f : X \to Y$ is **injective** if $\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$

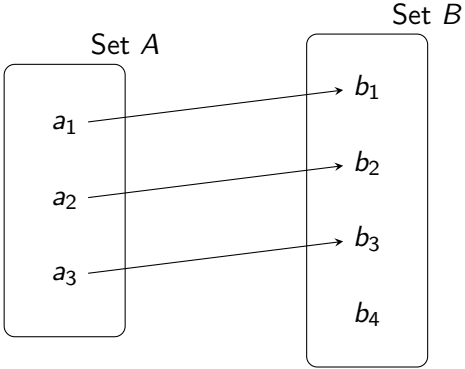
Monomorphism

A morphism $f : X \to Y$ in a category $\mathscr{C}$ is a **monomorphism** if for all objects Z and morphisms $g_1, g_2 : Z \to X$,

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2.$$

- In the category **Set**, monomorphisms coincide with injective functions.
- This definition generalizes injectivity to arbitrary categories.

Injectivity via Arrows



Universal Constructions

Definition

Universal Constructions are a way of defining properties up to Isomorphism

Example of Universal Construction: Terminal Object

Definition

An object T in a category $\mathscr{C}$ is *terminal* if for every object $X \in \mathrm{Ob}(\mathscr{C})$ there exists a unique morphism

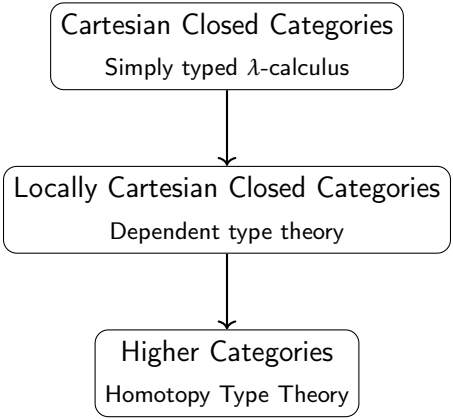
$$!_X : X \longrightarrow T$$

Take terminal objects $A, B \in \mathrm{Ob}(\mathscr{C})$. Then,

$$\begin{tikzcd}
b \arrow[dr] & & 1 \arrow[dl] \\
c \arrow[r] & A \arrow[r, leftrightarrow] & B & 2 \arrow[l] \\
d \arrow[ur] & & 3 \arrow[ul]
\end{tikzcd}$$

$$A \cong B$$

Perspectives



Errata

p.31 (Adgda -> Agda)

p.31 (Duplicated definition + Empty definition)

p.38 (Ax -> Assume-X)

p.48 (belements -> elements)

Inconsistencies here and there + Formatting issues

Q&A.

END.